Analysis Tutorial 8

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1 Introduction

In this tutorial we will consider density of irrational and rational numbers, Archimedean property, and we will solve problems related to these topics.

Definitions: Supremum and Infimum

Definition (Upper Bound and Lower Bound)

Let $A \subseteq \mathbb{R}$ be a non-empty set.

- A number $u \in \mathbb{R}$ is called an **upper bound** of A if

$$\forall x \in A,\ x \leq u.$$

- A number $l \in \mathbb{R}$ is called a **lower bound** of A if

$$\forall x \in A,\ l \leq x.$$

Definition (Supremum)

Let $A \subseteq \mathbb{R}$ be non-empty and bounded above.
A number $s \in \mathbb{R}$ is called the **supremum** (or **least upper bound**) of A if:

1. s is an upper bound of A, i.e.
$$\forall x \in A,\ x \leq s,$$
2. and for every $\varepsilon > 0$, there exists $x \in A$ such that
$$s - \varepsilon < x.$$

We denote this by $\sup A = s$.

Definition (Infimum)

Let $A \subseteq \mathbb{R}$ be non-empty and bounded below.
A number $t \in \mathbb{R}$ is called the **infimum** (or **greatest lower bound**) of A if:

1. t is a lower bound of A, i.e.
$$\forall x \in A,\ t \leq x,$$
2. and for every $\varepsilon > 0$, there exists $x \in A$ such that
$$x < t + \varepsilon.$$

We denote this by $\inf A = t$.

Problems:

Problem 1

If $a \in \mathbb{R}$ is such that

$$0 \leq a < \varepsilon \quad \text{for every } \varepsilon > 0,$$

then show that $a = 0$.

Problem 2

(Uniqueness of Least Upper Bound) Let E be a subset of $\mathbb{R}$. Then E can have at most one least upper bound.

Problem 3

Prove that every Cauchy sequence in $\mathbb{R}$ is bounded.

Problem 4

Let

$$A = \left\{ 1 - \frac{1}{n} : n \in \mathbb{N} \right\}.$$

Find $\sup A$, $\inf A$, $\max A$, and $\min A$ if they exist, and justify your answer.

Problem 5

Prove that if a sequence (x_n) in $\mathbb{R}$ satisfies $x_n \leq M$ for all n and $\lim_{n\to\infty} x_n = L$, then $L \leq M$.

Problem 6

Let

$$S = \{x \in \mathbb{R} : x < 0 \text{ or } x > 1\}.$$

(a) Find $\sup S$ and $\inf S$.

(b) Let $\mathbb{R} \setminus S$ denote the complement of S in $\mathbb{R}$. Find $\sup(\mathbb{R} \setminus S)$ and $\inf(\mathbb{R} \setminus S)$.

Problem 7

Let

$$C = \left\{ \frac{m}{n} : m, n \in \mathbb{N},\ n \geq 1,\ \frac{m}{n} < \sqrt{3} \right\}.$$

Show that $\sup C = \sqrt{3}$ even though $\sqrt{3} \notin C$.

Problem 8

Prove that if (x_n) is bounded and monotone, then (x_n) converges.

Problem 9

Let

$$E = \left\{ (-1)^n \left(1 - \frac{1}{n}\right) : n \in \mathbb{N} \right\}.$$

Find $\sup E$ and $\inf E$. *(Hint: consider separately the subsequences for even and odd n.)*

Problem 10

Let

$$S = \left\{ x \in \mathbb{R} : x^3 < 5 \right\}.$$

Prove rigorously (using the ε-definition of supremum) that

$$\sup S = \sqrt[3]{5}.$$

Problem 11

Let $S \subset \mathbb{R}$ be non-empty, bounded, and let $T = \{x^2 : x \in S\}$.

(a) Prove that $\sup T = (\sup |S|)^2$.

(b) Provide an example where $\sup T \neq (\sup S)^2$.